

Secretary Problem Research

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Lecture 1: abstract error SP

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1 Abstract Error SP

1.1 Basic Definitions

$i \approx j$ means that i and j are in-distinguishable in the first observation.

- (reflexivity) $i \approx i$
- (symmetry) $i \approx j \Leftrightarrow j \approx i$
- (connectedness) $i \approx j, r_i < r_j \Rightarrow \forall k, l [r_k, r_l \in [r_i, r_j] \Rightarrow r_k \approx r_l]$

1.2 Algorithmic Representations

Algorithm 1 Oct 14 2021

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1: for  $i \in [n]$  do
2:    $\alpha \leftarrow$  the current item to be decided
3:   if  $i < r$  then ▷ Phase 1
4:     if  $\exists \beta \in O_i. \alpha <_i \beta$  then ▷  $\alpha$  is not a maximum
5:       Decline this item
6:     else if  $\alpha \in E_i$  then
7:       Decline this item
8:     else
9:       Examine this item
10:    end if
11:  else if  $i < n$  then ▷ Phase 2
12:    if  $\exists \beta \in O_i. \alpha <_i \beta$  then
13:      Decline this item
14:    else
15:      if  $\forall \beta \in O_i. \beta \leq_i \alpha$  then ▷  $\alpha$  is the absolute maximum
16:        Accept this item
17:      else if  $\alpha \in E_i$  then
18:        Decline this item
19:      else
20:        Examine this item
21:      end if
22:    end if
23:  else
24:    Accept this item
25:  end if
26: end for
```

Algorithm 2 Game Process, Relaxed

```

1:  $[n], r_i$  and  $\approx$  relation are given according to definition
2:  $O_1 = \{1\}$ 
3:  $E_1 = \emptyset$ 
4:  $\alpha_1 = 1$ 
5:  $\Theta_1 = \emptyset$ 
6: for  $i \in [n]$  do
7:    $d_i \leftarrow$  player's decision
8:   if  $d_i = \text{Accept}$  then
9:     Game terminates with reward  $v(\alpha_i)$ 
10:  else if  $d_i = \text{Decline}$  then
11:     $O_{i+1} \leftarrow O_i \cup \{i+1\}$ 
12:     $E_{i+1} \leftarrow E_i$ 
13:     $\alpha_{i+1} \leftarrow i+1$ 
14:  else if  $d_i = \text{Examine}$  then
15:     $O_{i+1} \leftarrow O_i$ 
16:     $E_{i+1} \leftarrow E_i \cup \{i\}$ 
17:     $\alpha_{i+1} \leftarrow \alpha$ 
18:  end if
19:   $\Theta_{i+1} \leftarrow \{(\beta, \gamma) \in O_{i+1}^2 : r_\beta \leq r_\gamma, \beta \not\approx \gamma\} \cup \{(\beta, \gamma) \in E_{i+1}^2 : r_\beta \leq r_\gamma\}$ 
20: end for

```

Lecture 2: Relaxed SP

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2 Relaxed SP Prophet Inequality**2.1 classical SP PI**

We have n items with weights $X_1, \dots, X_n \sim \text{Unif}(0, 1)$. We adopt a thresholding scheme for ALG: We pick the first item encountered with weight $\geq T$.

$$\begin{aligned}
\mathbb{P}[X_{\max} \text{ selected}] &= \sum_{i=1}^n \mathbb{P}[X_1, \dots, X_{i-1} < T \cap X_i \geq T \cap X_i \text{ is max}] \\
&= \sum_{i=1}^n \int_0^1 \mathbb{P}[X_1, \dots, X_{i-1} < T \cap X_i \geq T \cap X_{i+1}, \dots, X_n \leq X_i \mid X_i = x] f_{\text{Unif}(0,1)}(x) dx \\
&= \sum_{i=1}^n \int_T^1 \mathbb{P}[X_1, \dots, X_{i-1} < T \cap X_{i+1}, \dots, X_n \leq x] f_{\text{Unif}(0,1)}(x) dx \\
&= \sum_{i=1}^n \int_T^1 T^{i-1} x^{n-i} dx \\
&= \sum_{i=1}^n \frac{1}{n-i+1} T^{i-1} (1 - T^{n-i+1})
\end{aligned}$$

2.2 relaxed model PI

2.2.1 calculation 1 (in meeting)

$$\begin{aligned}
& \sum_{i=1}^n \mathbb{P}[\text{accept } i \cap i \text{ best}] \\
&= \sum_{i=1}^n \mathbb{P}[\text{accept } i \mid i \text{ best}] \mathbb{P}[i \text{ best}] \\
&= \frac{1}{n} \sum_{i=1}^n \mathbb{P}[X_1, \dots, X_{i-1} < T \cap X_i \geq T \mid i \text{ best}] \\
&= \frac{1}{n} \sum_{i=1}^n \mathbb{P}[X_1, \dots, X_{i-1} < T \mid X_i \geq T \cap i \text{ best}] \mathbb{P}[X_i \geq T \mid i \text{ is best}]
\end{aligned}$$

Note that $\mathbb{P}[X_i \geq T \mid i \text{ best}] = 1 - F_{X_{\max}}(T) = 1 - T^n$. Also,

$$\begin{aligned}
& \mathbb{P}[X_1, \dots, X_{i-1} < T \mid X_i \geq T \cap i \text{ best}] \\
&= \int_T^1 \mathbb{P}[X_1, \dots, X_{i-1} < T \mid i \text{ best} \cap X_i = x] f_X(x \mid X_i \geq T \cap i \text{ best}) dx \\
&= \int_T^1 \mathbb{P}[X_1, \dots, X_{i-1} < T \mid X_1, \dots, X_{i-1} < x] f_{X_{\max}}(x \mid X_{\max} \geq T) dx \\
&= \int_T^1 \left(\frac{T}{x}\right)^{i-1} \frac{nx^{n-1}}{1-T^n} dx \\
&= \frac{nT^{i-1}}{1-T^n} \int_T^1 x^{n-i} dx \\
&= \frac{nT^{i-1}}{1-T^n} \frac{1-T^{n-i+1}}{n-i+1}
\end{aligned}$$

Combining all terms,

$$\begin{aligned}
& \sum_{i=1}^n \mathbb{P}[\text{accept } i \cap i \text{ best}] \\
&= \frac{1}{n} \sum_{i=1}^n \frac{nT^{i-1}}{1-T^n} \frac{x^{n-i+1}}{n-i+1} (1-T^n) \\
&= \sum_{i=1}^n T^{i-1} \frac{1-T^{n-i+1}}{n-i+1}
\end{aligned}$$

2.2.2 Calculation 2

We have n items with weights $X_1, \dots, X_n \sim \text{Unif}(0, 1)$. Observed values are $\tilde{X}_i \sim \text{Unif}(X_i - \delta, X_i + \delta)$. We adopt a thresholding scheme for ALG: We examine an item if $\tilde{x}_i > T$ and accept if $x_i > T$.

$$\begin{aligned}
\mathbb{P}[X_{\max} \text{ selected}] &= \sum_{i=1}^n \mathbb{P}[X_1, \dots, X_{i-1} < T \cap X_i \geq T \cap X_i \text{ is max}] \\
&= \sum_{i=1}^n \int_0^1 \mathbb{P}[X_1, \dots, X_{i-1} < T \cap X_i \geq T \cap X_{i+1}, \dots, X_n \leq X_i \mid X_i = x] f_{\text{Unif}(0,1)}(x) dx \\
&= \sum_{i=1}^n \int_T^1 \mathbb{P}[X_1, \dots, X_{i-1} < T \cap X_{i+1}, \dots, X_n \leq x] f_{\text{Unif}(0,1)}(x) dx \\
&= \sum_{i=1}^n \int_T^1 T^{i-1} x^{n-i} dx \\
&= \sum_{i=1}^n \frac{1}{n-i+1} T^{i-1} (1 - T^{n-i+1})
\end{aligned}$$